3008ICT Lab 9 (Pre-training, BERT, and Tokenization)

Questions and Discussions

Part 1: True / False Questions

1. The "Bitter Lesson" in AI suggests that complex, hand-engineered approaches will always outperform simple algorithms in the long run.

True / False?

2. When evaluating a language model, a lower perplexity score indicates better performance.

True / False?

3. BERT is fundamentally an autoregressive model capable of naturally generating text left-to-right.

True / False?

4. Modern pre-trained language models rely heavily on subword tokenisation instead of standard word-level embeddings like GloVe.

True / False?

Part 2: Multiple Choice Questions

5. Which of the following best describes how BERT achieves bidirectionality?

a. It concatenates two independent unidirectional LSTMs (forward and backward).
b. It uses a standard autoregressive objective but reads the text from right-to-left.
c. It utilises the Transformer architecture and a Masked Language Modeling (MLM) objective.
d. It uses Byte Pair Encoding to reverse the order of characters.

6. What is the primary purpose of the [CLS] token in BERT?

a. To act as a boundary separating two different sentences.
b. To predict the masked words in the sequence.
c. To represent the aggregated sequence-level information for classification tasks.
d. To indicate the end of a generated sequence.

7. How does BERT handle extractive Question Answering tasks (like SQuAD)?

a. It generates the answer sequence left-to-right.
b. It predicts the start and end indices of the answer span within the provided passage.
c. It classifies the entire passage as either 'True' or 'False'.
d. It retrieves the answer from an external knowledge base.

8. Why did modern NLP models move away from strictly character-level models?

a. Character-level models cannot handle out-of-vocabulary terms.
b. They resulted in sequences that were too long, making computation extremely expensive.
c. They required massive softmax matrices that slowed down training.
d. They overfit easily on small datasets.

Part 3: Short & Long Answer Questions

9. Explain the mechanism of Byte Pair Encoding (BPE). How does it construct its vocabulary?

10. Contrast the bidirectionality of ELMO with the bidirectionality of BERT. Why is BERT's approach considered "deeply bidirectional”?